Optimizitation and Validation for Okafor Reduced Reaction Mechanism

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Research Background

Traditional sources of energy rely on combustion of fossil fuels that produce carbon emissions which have adverse effects on the environment:

- Global Warming
- Climate Change

New carbon-free alternative fuels are needed to prevent exacerbating the effects of carbon pollutants

- The Japanese government seeks to reduce total greenhouse gas emissions by 46
- By 2050, Japan envisions its cities to be carbon-neutral

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Ammonia, NH_3, is a compound that interests researchers and scientists as it presents its as a potential solution to our energy needs in a variety of applications.

One of such applications is burning ammonia. Ammonia is a viable fuel for applications where combustion is most apt because:

- Ammonia combustion produces no CO_2
- Ammonia has a high energy density — both gravimetric and volumetry energy densities
- Easier to store than hydrogen
- Infrastructure to produce and transport ammonia already exists

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Compared to traditional fossil fuels and other alternative fuels, ammonia has some demerits:

- Laminar burning velocity of ammonia is slower than methane (CH_4) and hydrogen (H_2)
- Ammonia has a narrow flammability range
- Produces NO and NO_2 as main output

To better study ammonia combustion to further analyze the properties of the flame numerical simulations are used. For more reliable simulated data, accurate reaction mechanisms are required.

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Abe Mechanism is developed based on Okafor Reduced Reaction Mechanism[1] to improve laminar burning velocity predictions for NH_3/H_2/Air flames.

- Introduces N_2H_x reaction pathways for more detailed $NH_2 \rightarrow NH$ reactions
- Recognises the importance of $NNH \leftrightarrow N_2 + H$ in reaction zone of the flame
- Updated constants for Arrhenius function for reactions

[1] Ekenechukwu Chijioke Okafor et al. "Measurement and Modelling of the Laminar Burning Velocity of Methane-Ammonia-Air Flames at High Pressures Using a Reduced Reaction Mechanism". In: *Combustion and Flame* 204 (June 1, 2019), pp. 162–175. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2019.03.008.

- Numerically simulate flames with Abe mechanism at pressures 1, 3 and 5 atm for lean, stoichiometric and rich equivalence ratios.
  - Flame 1: $NH_3/H_2/Air$
  - Flame 2: $CH_4/NH_3/Air$

- Properties evaluated:
  - Laminar burning velocity
  - Species profiles of NO, NH_3, CO_2
  - Intermediate species profiles of OH, NNH, NH

- Validate by comparing numerically simulated values to experimental data from literature

- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily

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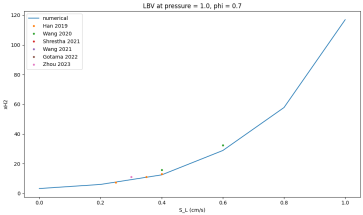
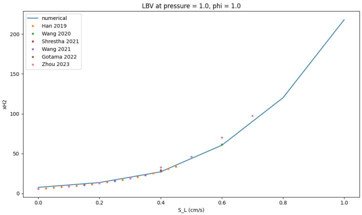
Validating Abe Mechanism

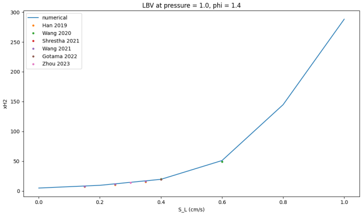
Laminar burning velocity of NH_3/H_2/Air flame at 1atm of pressure



(a) lean equivalence ratio

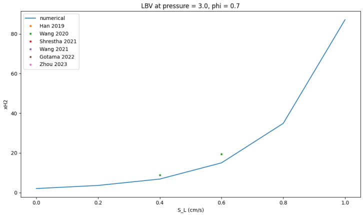


(b) stoichiometric equivalence ratio

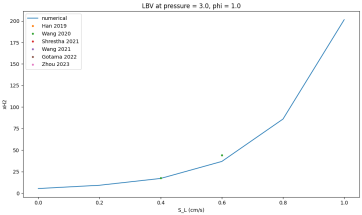


(c) rich equivalence ratio

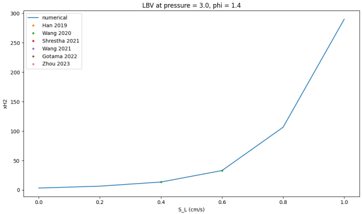
Laminar burning velocity of NH_3/H_2/Air flame at 3atm of pressure



(a) lean equivalence ratio

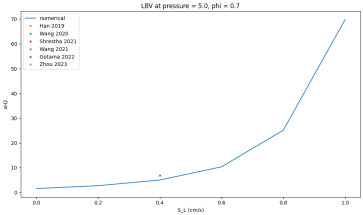


(b) stoichiometric equivalence ratio

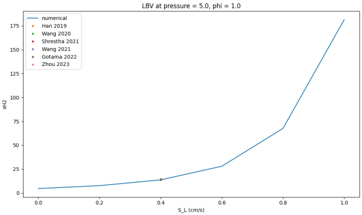


(c) rich equivalence ratio

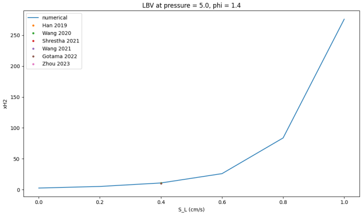
Laminar burning velocity of $NH_3/H_2/Air$ flame at 5atm of pressure



(a) lean equivalence ratio



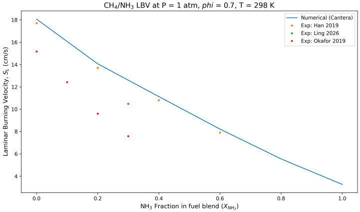
(b) stoichiometric equivalence ratio



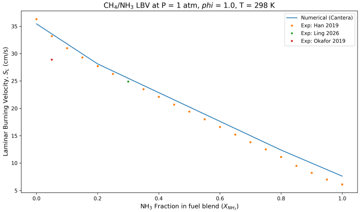
(c) rich equivalence ratio

Validating Abe Mechanism

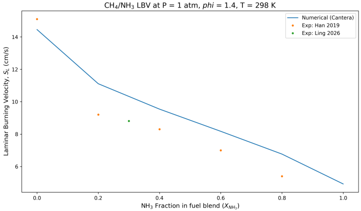
Laminar burning velocity of CH_4/NH_3/Air flame at 1atm of pressure



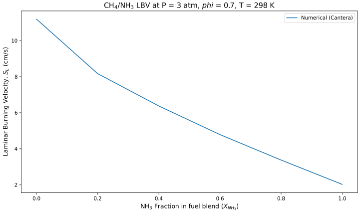
(a) lean equivalence ratio



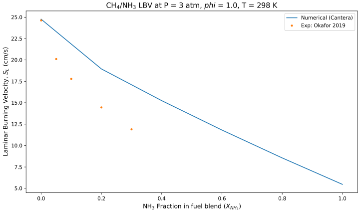
(b) stoichiometric equivalence ratio



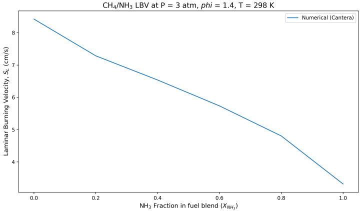
Laminar burning velocity of CH_4/NH_3/Air flame at 3atm of pressure



(a) lean equivalence ratio

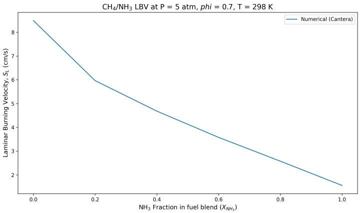


(b) stoichiometric equivalence ratio

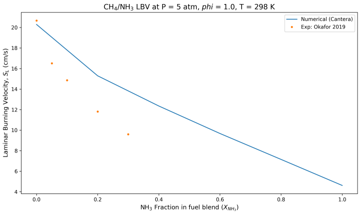


Validating Abe Mechanism

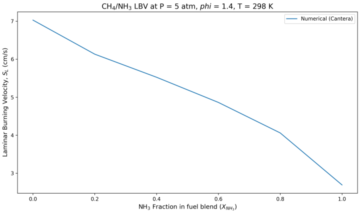
Laminar burning velocity of CH_4/NH_3/Air flame at 5atm of pressure



(a) lean equivalence ratio



(b) stoichiometric equivalence ratio



- Numerically evaluate species profiles with plug flow reactor
  - NO
  - NH_3
  - CO_2
- Numerically evaluate intermediate species profiles with stagnation flame model
  - OH
  - NNH
  - NH
- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily